MSG No. 85.59 $P4/n$ [Type I, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `-4..`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[1,4,6,7]`
2	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[2,3,5,8]`

Table 2: Wyckoff site: `2b`, site symmetry: `-4..`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[1,4,6,7]`
2	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[2,3,5,8]`

Table 3: Wyckoff site: `2c`, site symmetry: `4..`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[1,2,3,4]`
2	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	`[5,6,7,8]`

Table 4: Wyckoff site: `4d`, site symmetry: `-1`

No.	position	mapping
1	$[0, 0, 0]$	`[1,5]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[2,6]`
3	$\left[0, \frac{1}{2}, 0\right]$	`[3,7]`
4	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[4,8]`

Table 5: Wyckoff site: `4e`, site symmetry: `-1`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,5]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[2,6]`
3	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[3,7]`
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[4,8]`

Table 6: Wyckoff site: `4f`, site symmetry: `2..`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{3}{4}, z\right]$	`[1,4]`
2	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[2,3]`
3	$\left[\frac{3}{4}, \frac{1}{4}, -z\right]$	`[5,8]`
4	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	`[6,7]`

Table 7: Wyckoff site: 8g, site symmetry: 1

No.	position	mapping
1	$[x,\, y,\, z]$	[1]
2	$\left[\frac{1}{2}-y,\, x,\, z\right]$	[2]
3	$\left[y,\, \frac{1}{2}-x,\, z\right]$	[3]
4	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z\right]$	[4]
5	$[-x,\, -y,\, -z]$	[5]
6	$\left[y+\frac{1}{2},\, -x,\, -z\right]$	[6]
7	$\left[-y,\, x+\frac{1}{2},\, -z\right]$	[7]
8	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, -z\right]$	[8]